Geometry-Preserving Text Embedding Distillation via Manifold Diffusion

Anonymous submission

Abstract

Text embedding models induce semantic geometries whose neighborhoods, paths, and low-frequency structures shape how representations transfer across tasks. Existing distillation objectives usually match teacher coordinates, token-level states, or mini-batch similarities. While useful, these signals provide local or coordinate-dependent supervision and may fail to preserve the intrinsic geometry induced by a stronger teacher.

We introduce **Manifold Diffusion Distillation (MDD)**, a geometry-preserving framework for text embedding distillation. MDD treats teacher embeddings as finite support samples of a teacher-induced semantic manifold, constructs a sparse neighborhood graph, and defines multi-scale Markov diffusion profiles over this graph. A compact student is trained to match these diffusion profiles over sampled neighborhoods, while a spectral manifold anchor preserves low-frequency topology through graph Laplacian coordinates. The method requires only final teacher embeddings and does not modify the student architecture.

We provide a finite-support analysis showing that MDD is a controlled surrogate for preserving teacher-induced diffusion geometry under candidate-mass and KL-matching conditions. Empirically, frozen-representation probes on classification, pair classification, and semantic textual similarity show that MDD transfers teacher geometry into compact embeddings across class-region separability, pairwise semantic discrimination, and continuous similarity calibration. On Qwen3-Embedding-4B $\rightarrow$ BERT-base, MDD improves the overall frozen-probe average from 77.60 to 78.04 over the strongest baseline, with larger gains on SciTail and STS-Benchmark.

1 Introduction

Text embedding models induce a geometry over natural language data. Similarity, clustering, semantic transfer, and retrieval behavior are downstream manifestations of this geometry: useful embeddings organize examples into neighborhoods, regions, and continuous similarity structures that reflect meaning. Strong recent embedding models, including LLM-based encoders, improve this organization but are costly to run at scale. This motivates compact student encoders that inherit the semantic structure of a stronger teacher without reproducing every teacher coordinate or internal state.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, and relational losses that match pairwise similarities within a mini-batch. These objectives are attractive because they are simple and efficient. They also fit naturally into stochastic training.

However, embedding models are evaluated through geometric properties that these losses only indirectly supervise. Class-level transfer depends on whether semantic regions remain separable, pairwise decisions depend on local boundaries, and similarity tasks depend on calibrated distances. A pointwise loss asks the student to reproduce teacher coordinates, even when a compact student may not have enough capacity to match the teacher's absolute geometry. A mini-batch relational loss improves on this by comparing local similarity patterns, but it sees only the examples that happen to co-occur in a batch. Many relations that matter for representation transfer are neither pointwise nor confined to a single batch: samples may be connected through multi-hop semantic paths, and useful regions often emerge from the accumulated geometry of the support set.

This mismatch motivates the central question of this work: can a compact student inherit the teacher-induced diffusion geometry of a semantic manifold, rather than only matching individual vectors or mini-batch relations? The intuition is simple. If the teacher's embedding space defines a semantic manifold over a support set, then random walks on a teacher-induced neighborhood graph reveal how semantic mass flows from each example to nearby examples, broader local regions, and higher-order communities.

Our contributions are:

- We formulate text embedding distillation as preserving the teacher-induced semantic manifold and introduce Manifold Diffusion Distillation, which transfers multi-scale Markov diffusion profiles and low-frequency spectral structure into a compact student encoder.
- We provide finite-support analysis connecting the MDD

objective to diffusion-geometry preservation, including multi-scale barycenter matching, candidate-truncation error, low-frequency topology preservation, and the coverage gap of batch-local relational objectives.

- We evaluate the method through geometry-sensitive frozen embedding probes covering class-region separability, pairwise semantic discrimination, and continuous similarity calibration.

MDD first caches teacher embeddings for an unlabeled support set and constructs a mutual k-nearest-neighbor graph using teacher cosine similarity. From this graph, it defines a Markov transition matrix and precomputes diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

MDD adds two stabilizing signals. First, a spectral manifold objective projects student embeddings into low-dimensional spectral coordinates computed from the teacher graph Laplacian, encouraging the student to preserve global semantic topology rather than only local neighbors. Second, a lightweight pointwise anchor aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. A simple curriculum introduces diffusion scales gradually: the student first learns local neighborhoods, then semantic regions, and finally longer-range diffusion with spectral anchoring.

2 Related Work

Embedding Distillation Beyond Coordinate Matching. Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders by matching logits, hidden states, attention distributions, or self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). These methods provide effective compression recipes, but many rely on teacher internals, token-level correspondences, or output distributions that are not always available for black-box embedding teachers.

Embedding-specific distillation usually supervises final representations or task-derived relations. Sentence embedding methods such as DistilCSE transfer contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Gecko uses LLM-generated and LLM-relabelled retrieval data, while Jasper and Stella combine multi-stage embedding distillation with Matryoshka Representation Learning (Lee et al. 2024; Zhang et al. 2024a). LEAF supports asymmetric retrieval by aligning compact students to teacher representations, and Jina Embeddings v5 combines task-targeted distillation with contrastive training for compact multilingual embeddings (Vujanic and Rueckstiess 2025; Akram et al. 2026). These approaches establish distillation as a central tool for embedding model development. MDD differs in the object being transferred: instead of matching individual coordinates, generated pairs, or mini-batch similarities, it matches multi-scale diffusion profiles induced by the teacher geometry.

Diffusion Geometry and Manifold Learning. Graph-based manifold learning studies how local neighborhoods reveal low-dimensional structure in high-dimensional data. Laplacian Eigenmaps preserve local neighborhoods through graph Laplacian eigenvectors (Belkin and Niyogi 2003), while Diffusion Maps use Markov diffusion on a neighborhood graph to expose multi-scale connectivity and intrinsic geometry (Coifman and Lafon 2006). In this view, the important object is not an arbitrary coordinate system but the diffusion operator, its neighborhood expansion behavior, and the low-frequency modes of the graph.

MDD adapts this perspective to text embedding distillation. The teacher embeddings define finite support samples from a semantic manifold, and a sparse mutual kNN graph provides a discrete approximation of that manifold. Powers of the Markov transition matrix define diffusion profiles that describe how semantic mass spreads from an anchor to local neighbors, broader regions, and higher-order communities. The spectral anchor complements this local diffusion supervision by transferring low-frequency Laplacian coordinates. Thus, the student is trained to preserve teacher-induced diffusion geometry rather than only reproduce teacher coordinates.

Geometry-Preserving Representation Transfer. Relational knowledge distillation argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this often becomes mini-batch similarity matching, where teacher and student pairwise similarity matrices are compared over the examples sampled in a batch. This improves over pure coordinate regression, but the supervision remains local to the sampled batch and does not explicitly model multi-hop semantic paths or broader community structure.

Other representation alignment tools, including CKA-style similarity analysis, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful ways to compare or transfer representations (Kornblith et al. 2019; Alqahtani et al. 2021; Truong et al. 2025; Dao et al. 2026). These methods are valuable when intermediate layers or alignment pairs are available, but they generally optimize local representational agreement rather than a coordinate-free manifold object. MDD keeps the architecture-agnostic advantage of final-embedding distillation while replacing batch-local relational snapshots with cached diffusion targets sampled from the teacher-induced support graph.

Text Embedding Evaluation as Probes. Text embedding models are commonly evaluated through retrieval, classification, clustering, semantic textual similarity, and transfer benchmarks. Sentence-BERT and SimCSE made sentence-level similarity evaluation central to embedding research (Reimers and Gurevych 2019; Gao, Yao, and Chen

2021), while BEIR and MTEB broadened evaluation toward multi-task retrieval and representation assessment (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible embedding evaluation as models and datasets evolve (Chung et al. 2025). Recent large embedding systems such as BGE-M3, Qwen3 Embedding, and Dewey provide strong teacher candidates and broad deployment motivation (Chen et al. 2024; Zhang et al. 2025; Zhang, Zou, and Zhou 2025).

Our experiments use classification, pair classification, and semantic textual similarity as frozen-representation probes rather than as claims of exhaustive downstream coverage. Classification probes test whether class-level semantic regions remain linearly separable; pair classification probes test local semantic discrimination; and STS probes test continuous similarity calibration. This evaluation matches our central claim: preserving teacher-induced diffusion geometry should yield compact embeddings that transfer across complementary aspects of semantic representation quality.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote an unlabeled support set sampled from the text distribution. A teacher encoder T maps each input to a cached teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \tag{1}$$

and a compact student encoder S_θ maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \tag{2}$$

The objective is to train S_θ so that the student preserves the semantic organization induced by the teacher over this finite support, including nearest-neighbor rankings, local semantic regions, and higher-order community structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N} \sum_{i=1}^{N} \left(1 - \cos(W s_i, t_i)\right), \tag{3}$$

where W is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch B:

$$\mathcal{L}_{\text{rel}} = \left\| G_B^S - G_B^T \right\|_F^2, \tag{4}$$

where G_B^S and G_B^T are cosine-similarity matrices over examples in B. These objectives are useful local surrogates, but they do not directly supervise the multi-scale neighborhood geometry induced by the teacher.

3.2 Motivation

MDD starts from the observation that representation transfer depends on how probability mass is organized over a teacher-induced support graph, not only on the absolute coordinate of each embedding. A compact student may be unable to reproduce the teacher's full embedding map, yet it can still preserve useful semantic geometry if it matches the teacher's neighborhood distributions and global topology.

We therefore treat the teacher embedding space as a semantic graph and distill the random-walk behavior of this graph. For an anchor x_i, a one-hop walk captures direct semantic neighbors; multi-hop walks reveal broader regions connected through intermediate examples. This makes the distillation target a soft geometric object: how semantic mass should spread from an anchor to nearby candidates at different scales.

For a diffusion scale r, write the full teacher diffusion row as

$$q_{i,r}(j) = (P_r^T)_{ij}, \qquad j \in \mathcal{D}. \tag{5}$$

Since matching a distribution over all N examples is expensive, MDD samples a candidate set C_i and uses the conditional diffusion distribution on that set. Its captured mass is

$$\rho_{i,r}(C_i) = \sum_{j \in C_i} q_{i,r}(j). \tag{6}$$

The method is designed so that C_i contains high-diffusion neighbors, hard negatives, and random negatives, making the restricted target informative while keeping training tractable.

The rest of the section defines the two geometric transfer losses and the stabilizing anchor objective.

3.3 Multi-Scale Diffusion Loss

MDD first encodes the support set with the teacher and constructs a teacher-induced support graph. For each example x_i, let

$$\mathcal{N}_k^T(i) = \mathrm{TopK}_{j \neq i} \cos(t_i, t_j) \tag{7}$$

be its top-k teacher nearest neighbors. To reduce noisy one-directional edges and hubness, MDD uses a mutual kNN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \tag{8}$$

The weighted adjacency matrix is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where τ_g controls graph sharpness. Mutual kNN can produce isolated support points, so we add self-loops and use a lazy random walk. Let $\bar{W}^T = W^T + I$ and let $\bar{D}^T$ be its diagonal degree matrix. The transition matrix is

$$P^T = \frac{1}{2} I + \frac{1}{2} (\bar{D}^T)^{-1} \bar{W}^T. \tag{10}$$

This ensures every support point has non-zero degree and stabilizes multi-step diffusion. MDD then defines multi-scale diffusion targets

$$P_r^T = (P^T)^r, \qquad r \in \mathcal{R}, \tag{11}$$

where $\mathcal{R}$ is typically $\{1, 2, 4\}$. Small r preserves sharp local neighborhoods, while larger r captures semantic regions and multi-hop communities.

For each anchor x_i, MDD builds a single candidate set shared across diffusion scales:

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \tag{12}$$

where C_i^+ contains high-probability diffusion neighbors, C_i^h contains teacher hard negatives, and C_i^r contains random negatives. The positives are selected from the weighted multi-scale teacher diffusion score

$$\tilde{q}_i(j) = \frac{\sum_{r\in\mathcal{R}} \omega_r q_{i,r}(j)}{\sum_{r\in\mathcal{R}} \omega_r}, \tag{13}$$

as

$$C_i^+ = \mathrm{TopM}_{j\neq i}\, \tilde{q}_i(j). \tag{14}$$

Hard negatives are teacher-near examples outside the positive set,

$$C_i^h = \mathrm{TopH}_{j\notin C_i^+\cup\{i\}} \cos(t_i, t_j), \tag{15}$$

and random negatives are sampled uniformly from the remaining support points,

$$C_i^r \sim \mathrm{Uniform}\left(\mathcal{D} \setminus (C_i^+ \cup C_i^h \cup \{i\})\right). \tag{16}$$

Thus the same candidate set contains multi-scale diffusion positives, teacher-similar non-positive examples, and background support points. The teacher diffusion target at scale r is the diffusion mass restricted and renormalized over C_i:

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j'\in C_i}(P_r^T)_{ij'}}, \qquad j \in C_i. \tag{17}$$

Given an anchor embedding s_i and candidate embeddings $\{s_j : j \in C_i\}$, the student induces a distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j'\in C_i}\exp(\cos(s_i, s_{j'})/\tau_s)}, \qquad j \in C_i, \tag{18}$$

where τ_s is the student temperature. The diffusion loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|}\sum_{i\in B}\sum_{r\in\mathcal{R}_e} \omega_r D_{\mathrm{KL}}\left(p_{i,r}^T \| p_i^S\right), \tag{19}$$

where ω_r is the scale weight and $\mathcal{R}_e$ is the set of active scales at epoch e.

Although the teacher target varies with r, the student distribution p_i^S is intentionally shared across active scales. The weighted objective is equivalent to matching a multi-scale diffusion barycenter. Define

$$\tilde{p}_i^T(j) = \frac{\sum_{r\in\mathcal{R}_e}\omega_r p_{i,r}^T(j)}{\sum_{r\in\mathcal{R}_e}\omega_r}, \qquad j \in C_i. \tag{20}$$

Then

$$\sum_{r\in\mathcal{R}_e} \omega_r D_{\mathrm{KL}}\left(p_{i,r}^T \| p_i^S\right) = \Omega_e D_{\mathrm{KL}}\left(\tilde{p}_i^T \| p_i^S\right) + \text{const.}, \tag{21}$$

where $\Omega_e = \sum_{r\in\mathcal{R}_e}\omega_r$ and the constant does not depend on the student. Thus MDD trains the student to match one weighted multi-scale diffusion profile rather than forcing it to satisfy incompatible scale-specific distributions.

3.4 Spectral Manifold Anchoring

Diffusion matching transfers local and multi-hop neighborhoods, but a student can still fit candidate distributions while distorting broader community structure. MDD therefore computes spectral coordinates from the teacher graph. Let the symmetric normalized graph Laplacian be

$$L^T = I - (\bar{D}^T)^{-1/2}\bar{W}^T(\bar{D}^T)^{-1/2}. \tag{22}$$

Let $U^T \in \mathbb{R}^{N\times m}$ contain the first m non-trivial eigenvectors of L^T, and let u_i^T be the row corresponding to x_i. A learnable projection maps student embeddings to this spectral space:

$$z_i^S = W_{\text{spec}} s_i. \tag{23}$$

The spectral loss is

$$\mathcal{L}_{\text{spec}} = \frac{1}{|B|}\sum_{i\in B}\left\| z_i^S - u_i^T \right\|_2^2. \tag{24}$$

The first non-trivial eigenvectors of the normalized graph Laplacian are low-frequency modes of the teacher semantic graph. They vary smoothly over strongly connected regions and separate broader communities, following the graph-manifold principles behind Laplacian Eigenmaps and Diffusion Maps (Belkin and Niyogi 2003; Coifman and Lafon 2006). The spectral loss therefore anchors the student to the teacher's global topology.

3.5 Optimization Objectives

MDD includes a lightweight pointwise anchor for stability:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|}\sum_{i\in B}\left(1 - \cos(W_a s_i, t_i)\right), \tag{25}$$

where W_a maps student embeddings to the teacher dimension. This term acts as a boundary condition on the embedding map, while the diffusion and spectral objectives carry the main geometric supervision.

The final objective is

$$\begin{aligned}\mathcal{L}_{\text{MDD}} = \lambda_1\mathcal{L}_{\text{diff}} + \lambda_2\mathcal{L}_{\text{spec}} \\ + \lambda_3\mathcal{L}_{\text{anchor}}.\end{aligned} \tag{26}$$

We use a diffusion-scale curriculum: the first epoch uses one-step diffusion, the second epoch adds two-step diffusion, and later epochs add the full scale set and spectral anchoring. This avoids forcing long-range, smoother structure before the student has learned reliable local neighborhoods.

3.6 Theoretical Analysis

We provide a finite-support analysis showing that MDD is a controlled surrogate for preserving teacher-induced diffusion geometry. The analysis separates four effects: multi-scale aggregation, candidate truncation, low-frequency spectral anchoring, and the coverage gap of purely mini-batch relational matching.

Multi-scale diffusion as barycenter matching. For an anchor x_i, let $\alpha_r = \omega_r / \sum_{r' \in \mathcal{R}_e} \omega_{r'}$ be the normalized active scale weights and define the multi-scale diffusion barycenter

$$\bar{p}_i^T(j) = \sum_{r \in \mathcal{R}_e} \alpha_r p_{i,r}^T(j), \qquad j \in C_i. \tag{27}$$

Proposition 1. For any student distribution p_i^S over C_i,

$$\begin{aligned} &\sum_{r \in \mathcal{R}_e} \alpha_r D_{\mathrm{KL}}(p_{i,r}^T \| p_i^S) \\ &\quad = D_{\mathrm{KL}}(\bar{p}_i^T \| p_i^S) \\ &\qquad + \sum_{r \in \mathcal{R}_e} \alpha_r D_{\mathrm{KL}}(p_{i,r}^T \| \bar{p}_i^T). \end{aligned} \tag{28}$$

The second term is independent of the student. Thus, optimizing the multi-scale KL loss is equivalent to matching the student to the teacher's multi-scale diffusion barycenter, up to a constant measuring scale diversity among teacher neighborhoods.

Proof sketch. Expand each KL term as

$$\sum_j p_{i,r}^T(j) \log p_{i,r}^T(j) - \sum_j p_{i,r}^T(j) \log p_i^S(j). \tag{29}$$

Adding and subtracting the barycenter log-density gives the stated decomposition.

Candidate-truncated diffusion matching. Let $q_{i,r}$ be the full teacher diffusion row over $\mathcal{D}$, let

$$\rho_{i,r}(C_i) = \sum_{j \in C_i} q_{i,r}(j), \tag{30}$$

and let $q_{i,r}^C$ be the conditional distribution of $q_{i,r}$ on C_i. Let $\widehat{p}_i^S$ be the zero extension of p_i^S from C_i to $\mathcal{D}$.

Theorem 1. Assume $\rho_{i,r}(C_i) \geq 1 - \epsilon$ for all $r \in \mathcal{R}_e$. If

$$\sum_{r \in \mathcal{R}_e} \alpha_r D_{\mathrm{KL}}(q_{i,r}^C \| p_i^S) \leq \eta, \tag{31}$$

then

$$\sum_{r \in \mathcal{R}_e} \alpha_r \left\| q_{i,r} - \widehat{p}_i^S \right\|_1 \leq 2\epsilon + \sqrt{2\eta}. \tag{32}$$

Consequently, for any bounded probe function $f : \mathcal{D} \to [-1, 1]$,

$$\begin{aligned} &\sum_{r \in \mathcal{R}_e} \alpha_r \left| \mathbb{E}_{j \sim q_{i,r}}[f(j)] - \mathbb{E}_{j \sim p_i^S}[f(j)] \right| \\ &\quad \leq 2\epsilon + \sqrt{2\eta}. \end{aligned} \tag{33}$$

This separates the approximation error into candidate truncation error ϵ and student matching error η.

Proof sketch. For each scale,

$$\left\| q_{i,r} - \widehat{p}_i^S \right\|_1 \leq 2(1 - \rho_{i,r}(C_i)) + \left\| q_{i,r}^C - p_i^S \right\|_1. \tag{34}$$

Pinsker's inequality bounds the second term by

$$\sqrt{2 D_{\mathrm{KL}}(q_{i,r}^C \| p_i^S)}. \tag{35}$$

Averaging over scales and applying Jensen's inequality gives the result. The probe bound follows from $|f| \leq 1$.

Low-frequency topology preservation. Let u_i^T be the teacher spectral coordinate and $z_i^S = W_{\mathrm{spec}} s_i$ be the student spectral prediction. Suppose the support-average spectral loss satisfies

$$\frac{1}{N} \sum_{i=1}^{N} \left\| z_i^S - u_i^T \right\|_2^2 \leq \xi. \tag{36}$$

Define $d_{T,m}(i,j) = \|u_i^T - u_j^T\|_2$ and $d_{S,m}(i,j) = \|z_i^S - z_j^S\|_2$.

Theorem 2. If (i,j) is sampled uniformly from $\mathcal{D} \times \mathcal{D}$, then

$$\mathbb{E}_{i,j}\left[|d_{S,m}(i,j) - d_{T,m}(i,j)| \right] \leq 2\sqrt{\xi}. \tag{37}$$

More generally, if the spectral coordinates are multiplied by diagonal diffusion weights whose largest retained weight is $a_{\max}$, the bound becomes $2 a_{\max} \sqrt{\xi}$.

Proof sketch. The reverse triangle inequality gives

$$\begin{aligned} &|d_{S,m}(i,j) - d_{T,m}(i,j)| \\ &\quad \leq \left\| z_i^S - u_i^T \right\|_2 + \left\| z_j^S - u_j^T \right\|_2. \end{aligned} \tag{38}$$

Taking expectation and applying Jensen's inequality gives the result. This guarantee concerns preservation of teacher low-frequency coordinates encoded by the student; it does not claim that the student-induced graph recovers the full teacher spectrum.

Mini-batch relational coverage gap. Let B be a batch of size b containing anchor i, with the other $b - 1$ examples sampled uniformly without replacement from $\mathcal{D} \setminus \{i\}$. For any teacher diffusion row $q_{i,r}$,

$$\mathbb{E}_B \left[\sum_{j \in B \setminus \{i\}} q_{i,r}(j) \right] = \frac{b-1}{N-1} \left(1 - q_{i,r}(i)\right). \tag{39}$$

When $b \ll N$, a batch-local relational objective observes only a small expected fraction of the diffusion neighborhood around each anchor. MDD avoids this coverage bottleneck by constructing C_i from high-diffusion neighbors, teacher hard negatives, and random negatives.

4 Experiments

4.1 Experimental Setup

Benchmarks and metrics. We evaluate MDD with a self-contained frozen-representation protocol organized around three geometry-sensitive probe families. For text classification, we report macro-F1 on BANKING77 (Casanueva et al. 2020), Emotion (Saravia et al. 2018), and TweetEval-Sentiment (Barbieri et al. 2020); these probes test whether class-level semantic regions remain linearly separable. For pair classification, we report average precision (AP) on MRPC (Dolan and Brockett 2005), SciTail (Khot, Sabharwal, and Clark 2018), and WiC (Pilehvar and Camacho-Collados 2019); these probes test whether local pairwise semantic boundaries are preserved. For semantic textual similarity, we report Spearman correlation on SICK (Marelli et al. 2014), STS12 (Agirre et al. 2012), and STS-Benchmark (Cer et al. 2017); these probes test whether continuous similarity geometry remains calibrated. Emotion,

Algorithm 1 MDD Training

Require: Support set $\mathcal{D}$, teacher T, student S_θ, graph size k, diffusion scales $\mathcal{R}$

Cache teacher embeddings $t_i = T(x_i)$ for all $x_i \in \mathcal{D}$
Build a mutual kNN teacher graph W^T
Add self-loops and compute lazy transition matrix P^T
Precompute sparse diffusion candidates and targets $p^T_{i,r}$
Compute spectral coordinates u^T_i from the graph Laplacian
for each training epoch **do**
  Activate diffusion scales according to the curriculum
  for each mini-batch B **do**
    Encode anchors and candidate texts with S_θ
    Compute student candidate distributions p^S_i
    Compute $\mathcal{L}_{\mathrm{diff}}$, $\mathcal{L}_{\mathrm{spec}}$, and $\mathcal{L}_{\mathrm{anchor}}$
    Update θ by minimizing $\mathcal{L}_{\mathrm{MDD}}$
  end for
end for
return Distilled student encoder S_θ

WiC, and STS-Benchmark are treated as in-domain benchmarks because their unlabeled training texts are used to construct the support set; the remaining datasets are used only for out-of-domain transfer evaluation.

Models. We keep two teacher-student configurations: BGE-M3 (Chen et al. 2024) $\rightarrow$ BERT-base (Devlin et al. 2019) and Qwen3-Embedding-4B (Zhang et al. 2025) $\rightarrow$ BERT-base. Teacher embeddings are cached once before MDD training. After graph construction, the teacher model is removed from GPU memory; student training uses cached teacher embeddings, sparse diffusion targets, and spectral coordinates.

Training data. The unlabeled support set is constructed from the text fields of the in-domain training splits. For pairwise datasets, both sides of each pair are flattened into individual support examples before graph construction. MDD builds the teacher-induced support graph and trains without using downstream labels, task-specific heads, or evaluation labels.

Baselines. We compare MDD with the full teacher, the undistilled BERT-base student, SimCSE-unsup (Gao, Yao, and Chen 2021), cross-tokenizer distillation methods CDM (Chen et al. 2025) and DSKD (Zhang et al. 2024b), sentence embedding distillation methods Jasper and Stella (Zhang et al. 2024a) and DistillCSE (Gao et al. 2021; Xu et al. 2023), EMO (Truong et al. 2025), and TALAS (Dao et al. 2026). Component-level comparisons against pointwise anchoring and mini-batch relational matching are included in the ablation study.

Evaluation protocol. For classification, the encoder is frozen and a logistic regression classifier is trained on top of the sentence embeddings. For pair classification and STS, examples are embedded independently and compared by cosine similarity. We therefore use these benchmarks as representation probes rather than task-specific fine-tuning targets or direct retrieval benchmarks. We report per-task scores, the average over in-domain tasks, the average over out-of-domain tasks, and the overall average across all benchmarks.

Implementation details. Unless otherwise specified, MDD uses graph size $k = 50$, graph temperature $\tau_g = 0.1$, diffusion scales $\{1, 2, 4\}$, candidate size 64, 32 diffusion positives, 16 hard negatives, 16 random negatives, spectral dimension 16, and five training epochs. The scale weights are $(1.0, 0.5, 0.25)$. The loss weights are $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.05$.

4.2 Geometry-Sensitive Representation Probes

Since our goal is to preserve the geometry of the teacher embedding space rather than optimize a specific downstream task, we evaluate the student using frozen embedding probes. Classification measures whether semantic regions remain linearly separable. Pair classification measures whether local pairwise semantic boundaries are preserved. STS measures whether the continuous similarity geometry of the teacher space is retained. Together, these probes provide complementary evidence about whether manifold-level supervision transfers useful embedding geometry.

Tables 1 and 2 define the main reporting format for the two teacher-student pairs. We group tasks by probe type and separately report in-domain, out-of-domain, and overall averages. In the Qwen3-Embedding-4B $\rightarrow$ BERT-base setting, MDD achieves the best overall average under this frozen-embedding protocol, with particularly strong out-of-domain transfer and similarity-oriented performance. The gains are not uniform across all datasets, which suggests that manifold diffusion primarily improves global similarity geometry and transfer robustness rather than every local decision boundary.

4.3 Ablation Study

Table 3 isolates the contribution of each MDD component. The ablations separate the effect of multi-scale diffusion, spectral manifold anchoring, the pointwise anchor, curriculum scheduling, and candidate composition. This distinguishes manifold diffusion transfer from simpler local alignment objectives.

Table 3: Ablation template for MDD. Checkmarks indicate enabled components. Avg-In, Avg-Out, and Avg-All follow the domain split in Tables 1 and 2.

Variant	$\mathcal{L}_{\mathrm{diff}}$	$\mathcal{L}_{\mathrm{spec}}$	$\mathcal{L}_{\mathrm{anchor}}$	Curr.	Full C_i	Avg-In	Avg-Out	Avg-All
MDD full	✓	✓	✓	✓	✓	–	–	–
w/o diffusion loss	–	✓	✓	✓	✓	–	–	–
w/o spectral anchoring	✓	–	✓	✓	✓	–	–	–
w/o anchor	✓	✓	–	✓	✓	–	–	–
w/o curriculum	✓	✓	✓	–	✓	–	–	–
w/o hard negatives	✓	✓	✓	✓	partial	–	–	–
w/o random negatives	✓	✓	✓	✓	partial	–	–	–
single scale $r = 1$	✓	✓	✓	–	✓	–	–	–
single scale $r = 4$	✓	✓	✓	–	✓	–	–	–
pointwise anchor only	–	–	✓	–	–	–	–	–
mini-batch relational	–	–	✓	–	–	–	–	–

Table 1: Frozen-embedding probe template for BGE-M3 → BERT-base. Classification tasks report macro-F1, pair classification tasks report average precision (AP), and STS tasks report Spearman correlation. Starred datasets are in-domain.

	Classification (F1)			**Pair Classification (AP)**			**STS (Spearman)**			**Domain Avg**		**Avg-All**
	Banking77	TweetEval	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STSB*	Avg-In	Avg-Out	Avg-All
BGE-M3 → BERT-base												
Teacher	–	–	–	–	–	–	–	–	–	–	–	–
Student base	–	–	–	–	–	–	–	–	–	–	–	–
SimCSE-unsup	–	–	–	–	–	–	–	–	–	–	–	–
CDM	–	–	–	–	–	–	–	–	–	–	–	–
DSKD	–	–	–	–	–	–	–	–	–	–	–	–
Jasper/Stella	–	–	–	–	–	–	–	–	–	–	–	–
DistillCSE	–	–	–	–	–	–	–	–	–	–	–	–
EMO	–	–	–	–	–	–	–	–	–	–	–	–
TALAS	–	–	–	–	–	–	–	–	–	–	–	–
MDD	–	–	–	–	–	–	–	–	–	–	–	–

Table 2: Frozen-embedding probe results for Qwen3-Embedding-4B → BERT-base. Classification tasks report macro-F1, pair classification tasks report average precision (AP), and STS tasks report Spearman correlation. Starred datasets are in-domain.

	Classification (F1)			**Pair Classification (AP)**			**STS (Spearman)**			**Domain Avg**		**Avg-All**
	Banking77	TweetEval	Emotion*	MRPC	SciTail	WiC*	SICK	STS12	STSB*	Avg-In	Avg-Out	Avg-All
Qwen3-Embedding-4B → BERT-base												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper/Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	92.34	73.74	64.41	85.97	82.66	68.65	78.38	72.27	79.98	71.01	80.89	77.60
MDD	**92.62**	72.27	66.42	83.57	**89.15**	66.43	**79.69**	70.43	**81.79**	**71.55**	**81.29**	**78.04**

5 Discussion

MDD is designed around the observation that embedding distillation should preserve the intrinsic geometry induced by the teacher, rather than only its coordinates or mini-batch similarity snapshots. Our evaluation does not aim to exhaust all downstream uses of text embeddings; instead, it tests whether preserving teacher-induced diffusion geometry yields a student representation that transfers across complementary semantic probes. The results suggest that manifold diffusion is most helpful for global similarity geometry and transfer robustness, while gains need not be uniform across every local decision boundary.

The method also has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that MDD requires a support-graph construction step. Building a teacher graph is worthwhile when the same support set enables substantial distillation, but it adds overhead compared with purely mini-batch objectives.

When MDD may fail. MDD depends on the quality and coverage of the teacher-induced support graph. If teacher embeddings contain spurious neighborhoods, diffusion can propagate those errors; if the support set is mismatched to the deployment distribution, the learned geometry may preserve the wrong regions. Highly disconnected graphs can limit useful diffusion even with self-loops, while overly long diffusion scales can oversmooth local distinctions that matter for pairwise decisions. The method also inherits the cost of graph construction and diffusion-target preprocessing, which may be expensive for rapidly changing or very large support sets. Future work can study approximate nearest-neighbor construction, adaptive diffusion scales, graph diagnostics for noisy teachers, and online graph refresh for changing support distributions.

6 Conclusion

We introduced MDD, a geometry-preserving distillation framework for compact text embedding models. Instead of matching only teacher vectors, token states, or mini-batch similarity snapshots, MDD distills multi-scale semantic diffusion over a teacher-induced mutual kNN graph and anchors low-frequency topology with spectral coordinates. This targets the semantic geometry that frozen embedding probes inspect: class-level regions, pairwise boundaries, and continuous similarity structure. Together, the diffusion, spectral, and anchor objectives position MDD as a practical alternative to local embedding distillation objectives when the goal is to transfer teacher-induced representation geometry.

References

Agirre, E.; Cer, D.; Diab, M.; and Gonzalez-Agirre, A. 2012. SemEval-2012 Task 6: A Pilot on Semantic Textual Similarity. In *Proceedings of the First Joint Conference on Lexical and Computational Semantics*, 385–393.

Akram, M. K.; Sturua, S.; Havriushenko, N.; Herreros, Q.; Günther, M.; Werk, M.; and Xiao, H. 2026. jina-embeddings-v5-text: Task-Targeted Embedding Distillation. *arXiv preprint arXiv:2602.15547*.

Alqahtani, S.; Lalwani, G.; Zhang, Y.; Romeo, S.; and Mansour, S. 2021. Using Optimal Transport as Alignment Objective for Fine-tuning Multilingual Contextualized Embeddings. *arXiv preprint arXiv:2110.02887*.

Barbieri, F.; Camacho-Collados, J.; Espinosa-Anke, L.; and Neves, L. 2020. TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 1644–1650.

Belkin, M.; and Niyogi, P. 2003. Laplacian Eigenmaps for Dimensionality Reduction and Data Representation. *Neural Computation*, 15(6): 1373–1396.

Casanueva, I.; Temcinas, T.; Gerz, D.; Henderson, M.; and Vulic, I. 2020. Efficient Intent Detection with Dual Sentence Encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, 38–45.

Cer, D.; Diab, M.; Agirre, E.; Lopez-Gazpio, I.; and Specia, L. 2017. SemEval-2017 Task 1: Semantic Textual Similarity Multilingual and Crosslingual Focused Evaluation. In *Proceedings of the 11th International Workshop on Semantic Evaluation*, 1–14.

Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; and Liu, Z. 2024. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *arXiv preprint arXiv:2402.03216*.

Chen, Y.; Liu, Y.; Meng, F.; Chen, Y.; Xu, J.; and Zhou, J. 2025. Enhancing Cross-Tokenizer Knowledge Distillation with Contextual Dynamical Mapping. *arXiv preprint arXiv:2502.11104*.

Chung, I.; Kerboua, I.; Kardos, M.; Solomatin, R.; and Enevoldsen, K. 2025. Maintaining MTEB: Towards Long Term Usability and Reproducibility of Embedding Benchmarks. *arXiv preprint arXiv:2506.21182*.

Coifman, R. R.; and Lafon, S. 2006. Diffusion Maps. *Applied and Computational Harmonic Analysis*, 21(1): 5–30.

Dao, Q. P.; Nguyen, H. S.; Pham, K. C.; Ngo Van, L.; Nguyen, D. T.-N.; Nguyen, T. H.; and Le, T. 2026. TALAS: Teacher-Anchored Layer Alignment with Adaptive Sharpness-Aware Minimization for Embedding Distillation. Manuscript.

Devlin, J.; Chang, M.-W.; Lee, K.; and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 4171–4186.

Dolan, W. B.; and Brockett, C. 2005. Automatically Constructing a Corpus of Sentential Paraphrases. In *Proceedings of the Third International Workshop on Paraphrasing*.

Gao, C.; Wu, X.; Wang, P.; Wang, J.; Zang, L.; Wang, Z.; and Hu, S. 2021. DistilCSE: Effective Knowledge Distillation for Contrastive Sentence Embeddings. *arXiv preprint arXiv:2112.05638*.

Gao, T.; Yao, X.; and Chen, D. 2021. SimCSE: Simple Contrastive Learning of Sentence Embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 6894–6910.

Hinton, G.; Vinyals, O.; and Dean, J. 2015. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*.

Jiao, X.; Yin, Y.; Shang, L.; Jiang, X.; Chen, X.; Li, L.; Wang, F.; and Liu, Q. 2020. TinyBERT: Distilling BERT for Natural Language Understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, 4163–4174.

Khot, T.; Sabharwal, A.; and Clark, P. 2018. SciTail: A Textual Entailment Dataset from Science Question Answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Kornblith, S.; Norouzi, M.; Lee, H.; and Hinton, G. 2019. Similarity of Neural Network Representations Revisited. In *Proceedings of the 36th International Conference on Machine Learning*, 3519–3529.

Lee, J.; Dai, Z.; Ren, X.; Chen, B.; Cer, D.; Cole, J. R.; Hui, K.; Boratko, M.; Kapadia, R.; Ding, W.; Luan, Y.; Duddu, S. M. K.; Abrego, G. H.; Shi, W.; Gupta, N.; Kusupati, A.; Jain, P.; Jonnalagadda, S. R.; Chang, M.-W.; and Naim, I. 2024. Gecko: Versatile Text Embeddings Distilled from Large Language Models. *arXiv preprint arXiv:2403.20327*.

Marelli, M.; Menini, S.; Baroni, M.; Bentivogli, L.; Bernardi, R.; and Zamparelli, R. 2014. A SICK cure for the evaluation of compositional distributional semantic models.

In *Proceedings of the 9th International Workshop on Semantic Evaluation*, 216–223.

Muennighoff, N.; Tazi, N.; Magne, L.; and Reimers, N. 2023. MTEB: Massive Text Embedding Benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, 2014–2037.

Park, W.; Kim, D.; Lu, Y.; and Cho, M. 2019. Relational Knowledge Distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 3967–3976.

Pilehvar, M. T.; and Camacho-Collados, J. 2019. WiC: The Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 1267–1273.

Reimers, N.; and Gurevych, I. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 3982–3992.

Sanh, V.; Debut, L.; Chaumond, J.; and Wolf, T. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

Saravia, E.; Liu, H.-C. T.; Huang, Y.-H.; Wu, J.; and Chen, Y.-S. 2018. CARER: Contextualized Affect Representations for Emotion Recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 368–379.

Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.

Truong, M.-P.; Vu, H. A.; Vu, T.; Diep, N. T. N.; Van, L. N.; Nguyen, T. H.; and Le, T. 2025. EMO: Embedding Model Distillation via Intra-Model Relation and Optimal Transport Alignments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 7594–7606.

Vujanic, R.; and Rueckstiess, T. 2025. LEAF: Knowledge Distillation of Text Embedding Models with Teacher-Aligned Representations. *arXiv preprint arXiv:2509.12539*.

Wang, W.; Wei, F.; Dong, L.; Bao, H.; Yang, N.; and Zhou, M. 2020. MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. In *Advances in Neural Information Processing Systems*.

Xu, J.; Shao, W.; Chen, L.; and Liu, L. 2023. DistillCSE: Distilled Contrastive Learning for Sentence Embeddings. *arXiv preprint arXiv:2310.13499*.

Xu, X.; Li, M.; Tao, C.; Shen, T.; Cheng, R.; Li, J.; Xu, C.; Tao, D.; and Zhou, T. 2024. A Survey on Knowledge Distillation of Large Language Models. *arXiv preprint arXiv:2402.13116*.

Zhang, D.; Li, J.; Zeng, Z.; and Wang, F. 2024a. Jasper and Stella: Distillation of SOTA Embedding Models. *arXiv preprint arXiv:2412.19048*.

Zhang, D.; Zou, P.; and Zhou, Y. 2025. Dewey Long Context Embedding Model: A Technical Report. *arXiv preprint arXiv:2503.20376*.

Zhang, S.; Zhang, X.; Sun, Z.; Chen, Y.; and Xu, J. 2024b. Dual-Space Knowledge Distillation for Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 18164–18181.

Zhang, Y.; Li, M.; Long, D.; Zhang, X.; Lin, H.; Yang, B.; Xie, P.; Yang, A.; Liu, D.; Lin, J.; Huang, F.; and Zhou, J. 2025. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. *arXiv preprint arXiv:2506.05176*.